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PostgreSQL's Rise to Power: Why the Open-Source Giant is Dominating the Database Landscape

Who am I:

Engineer and Advocate at Redgate

30 Years in the Tech Industry

DBA (Oracle, SQL Server, MySQL, PostgreSQL...)

DevOps, Analytics and AI

Cloud Migrations and High IO Infrastructure

Co-Leader for the MS Data Platform DEI Group

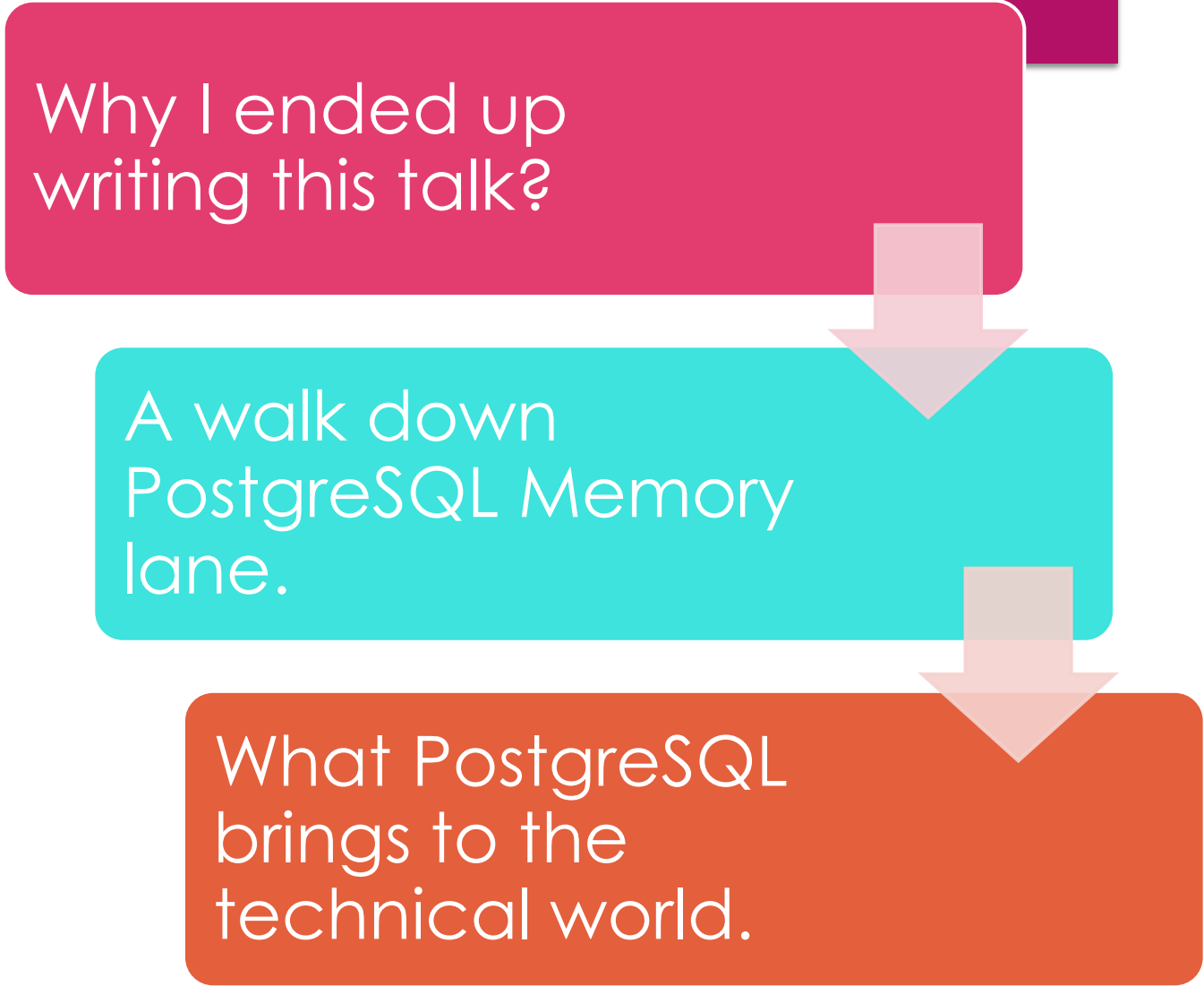
AI Advisor to Numerous Organizations



Kellyn Gorman

The 10K View

Why I ended up
writing this talk?



A walk down
PostgreSQL Memory
lane.

What PostgreSQL
brings to the
technical world.

Why Did I Write This Talk



I come from the **Enterprise Database** world....they like Mikey from Life cereal....



Lots of **myths** About PostgreSQL in the wild.



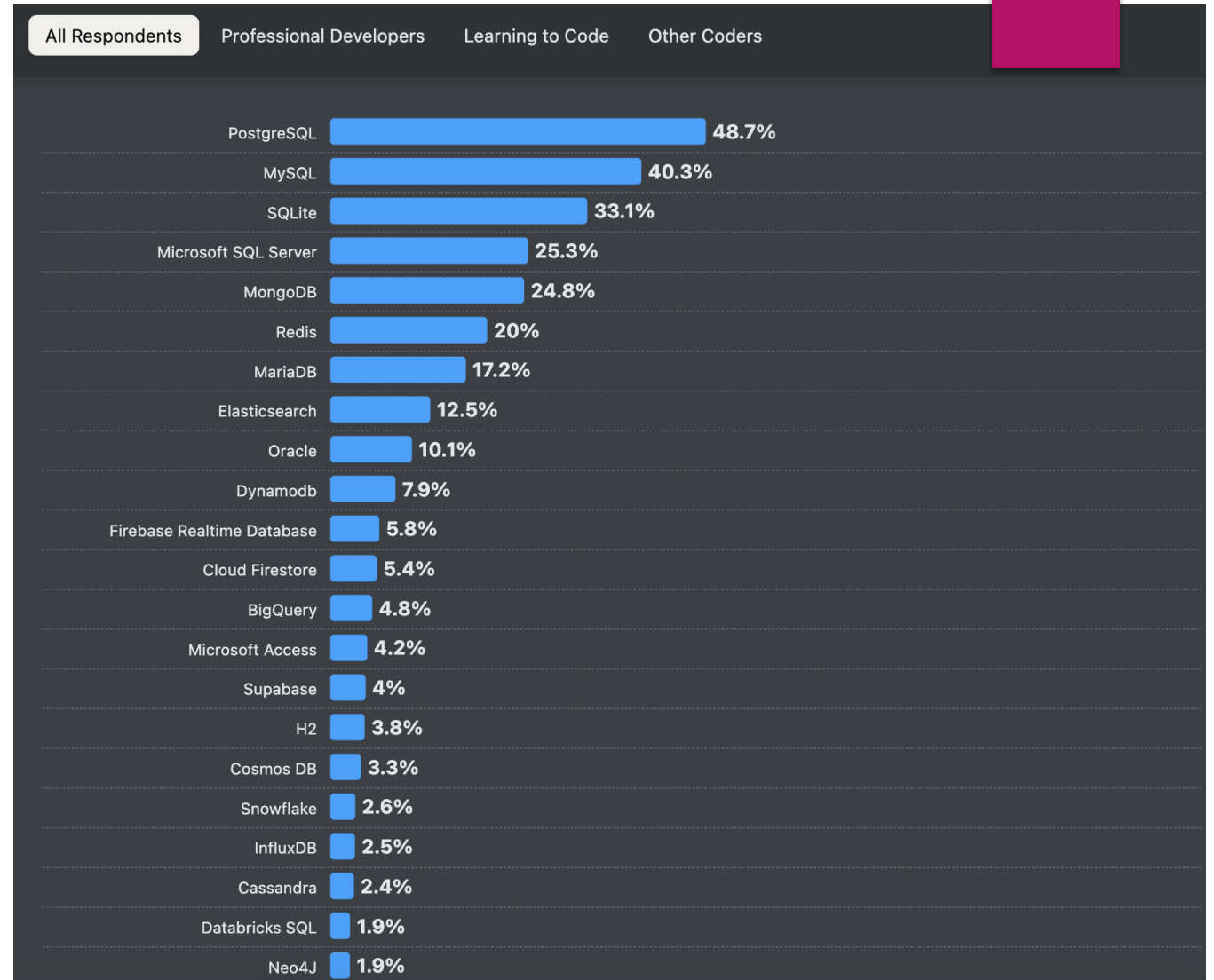
I wanted to understand the **big picture** – I was seeing the patterns.



It isn't **ALL** sunshine and lollipops though...

PostgreSQL by the Numbers

STACK OVERFLOW
DEVELOPERS SURVEY 2024:
POSTGRESQL OVERTOOK
MYSQL AS MOST LOVED &
WANTED RDBMS.



DB-Engines Ranking: Consistent Year-Over-Year rise

<https://db-engines.com/en/ranking>

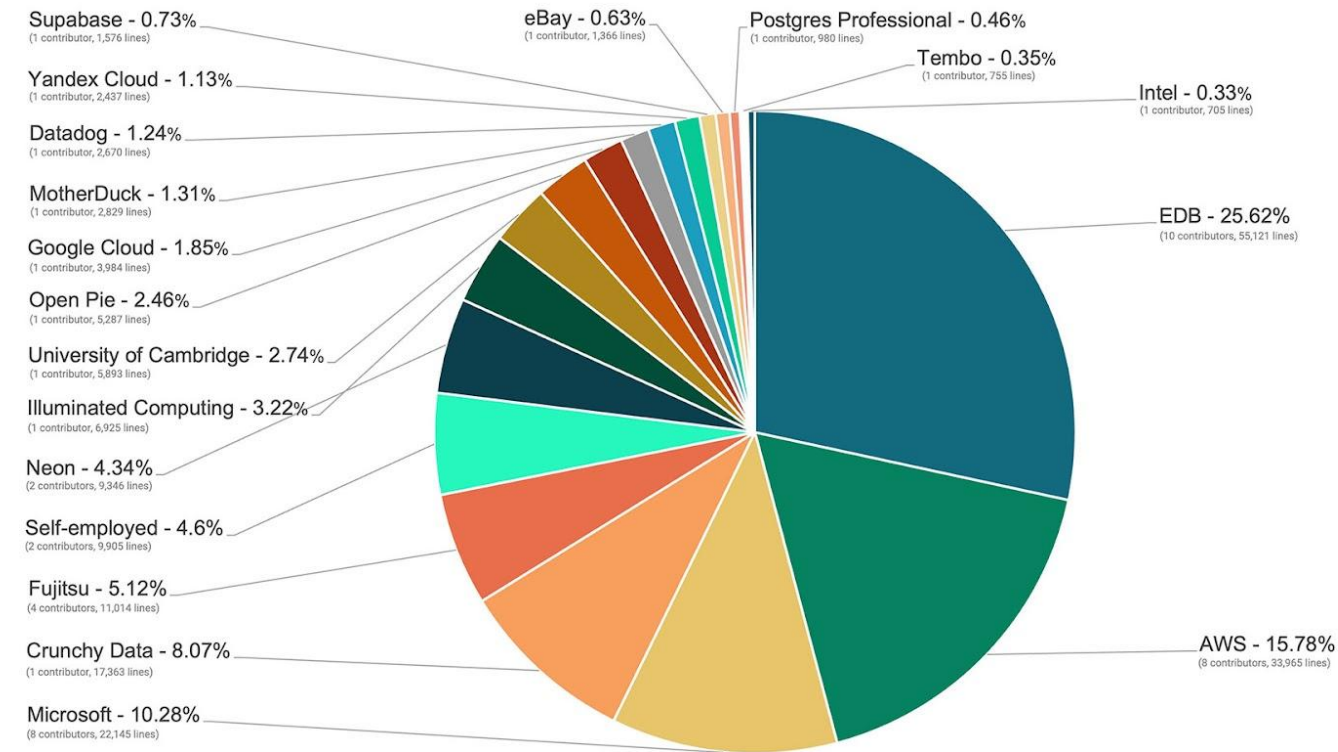
424 systems in ranking, July 2025

Rank			DBMS	Database Model	Score		
Jul 2025	Jun 2025	Jul 2024			Jul 2025	Jun 2025	Jul 2024
1.	1.	1.	Oracle	Relational, Multi-model ⓘ	1217.05	-13.33	-23.31
2.	2.	2.	MySQL	Relational, Multi-model ⓘ	940.73	-12.85	-98.73
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model ⓘ	771.14	-5.61	-36.51
4.	4.	4.	PostgreSQL +	Relational, Multi-model ⓘ	680.89	+0.23	+41.98
5.	5.	5.	MongoDB +	Document, Multi-model ⓘ	403.83	+0.99	-25.99
6.	6.	↑ 7.	Snowflake	Relational	176.17	+1.68	+39.64
7.	7.	↓ 6.	Redis	Key-value, Multi-model ⓘ	149.72	-2.01	-7.05
8.	8.	↑ 9.	IBM Db2	Relational, Multi-model ⓘ	127.51	+2.38	+3.11
9.	9.	↓ 8.	Elasticsearch	Multi-model ⓘ	118.83	-2.45	-12.00
10.	10.	10.	SQLite	Relational	115.44	-1.60	+5.49

Top 4 overall!

Companies that had the biggest impact on PostgreSQL development in 2024

The following companies, through their team members, contributed 90% of the new lines of code to Postgres 17



It's a Favorite of Cloud Vendors

- ▶ Every Hyperscaler has a PostgreSQL-based offering:
 - ▶ Amazon RDS for PostgreSQL
 - ▶ Azure Database for PostgreSQL
 - ▶ Flexible Server
 - ▶ Single Server, (not deprecated)
 - ▶ Google Cloud SQL for PostgreSQL
- ▶ Aiven for PostgreSQL
- ▶ Crunchy Bridge (Crunchy Data)

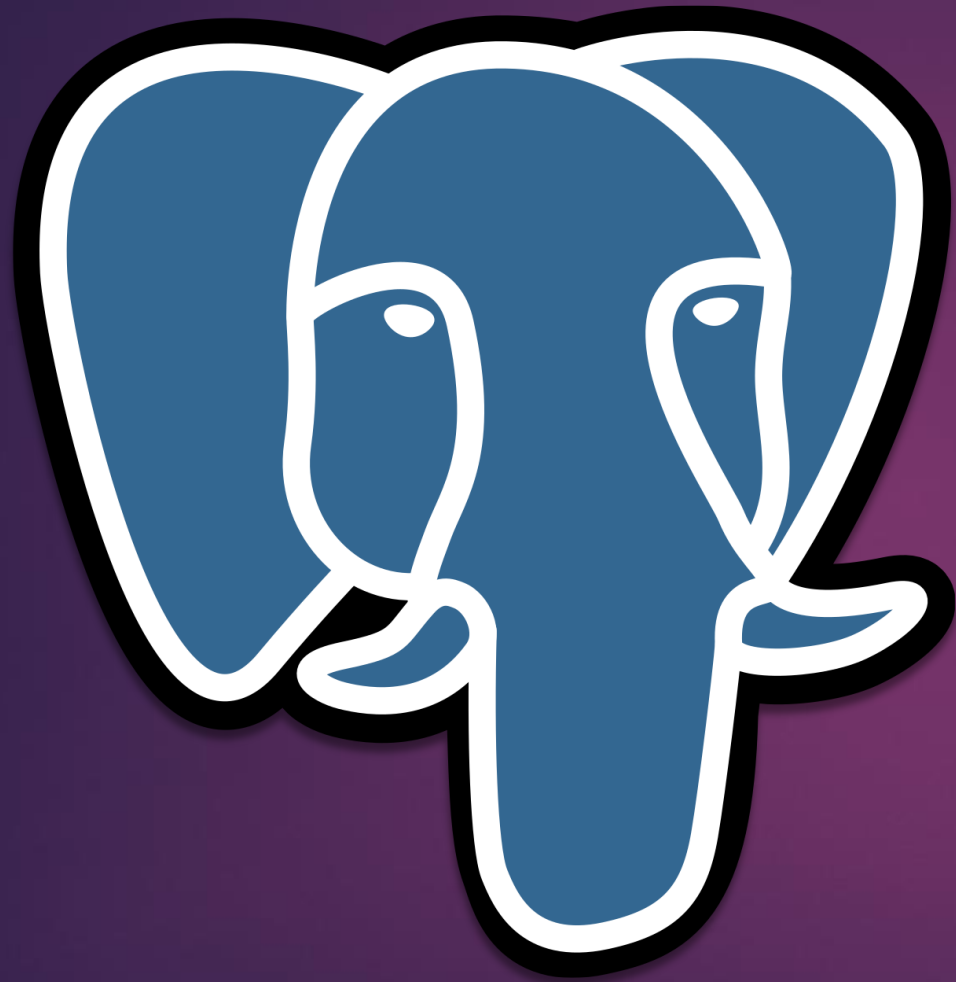
PostgreSQL-Compatible Databases

- ▶ Amazon Aurora PostgreSQL- Compatible
 - ▶ Comes with distributed storage, faster replication and improved failover.
- ▶ Google AlloyDB for PostgreSQL
- ▶ Google AlloyDB Omni, (can be decoupled from Google cloud)
- ▶ Microsoft Fabric Lakehouse with PostgreSQL front-end
- ▶ YugabyteDB
 - ▶ Distributed SQL database with PostgreSQL-compatible API (YSQL) at the storage layer.
- ▶ CockroachDB
 - ▶ Another distributed SQL database that speaks PostgreSQL wire protocol, but not completely compatible.

Additional Contenders

- ▶ Managed PostgreSQL
 - ▶ ScaleGrid
 - ▶ ElephantSQL
 - ▶ Supabase(core PostgreSQL)
 - ▶ Render(supports PostgreSQL as a Service)
 - ▶ Neon Serverless PostgreSQL with autoscaling and branching
 - ▶ Nile (serverless PostgreSQL platform made for multitenant SaaS)

Provider	Type	Compatibility	Notes
Amazon RDS	Managed PostgreSQL	Full	Native PostgreSQL engine
Amazon Aurora	Custom Engine	Compatible	Custom storage and enhancements
Azure PostgreSQL	Managed PostgreSQL	Full	Flexible Server recommended
GCP Cloud SQL	Managed PostgreSQL	Full	Native PostgreSQL
Google AlloyDB	Custom Engine	Compatible	Optimized for analytics/performance
Crunchy Bridge	Managed PostgreSQL	Full	Enterprise-focused
Aiven	Managed PostgreSQL	Full	Multi-cloud support
YugabyteDB	Distributed SQL	Compatible	Full SQL with some PostgreSQL compatibility
CockroachDB	Distributed SQL	Partial	PostgreSQL wire-compatible
Neon	Serverless PostgreSQL	Full	Modern branching and autoscaling features
Supabase	App Backend Platform	Full	Open-source Firebase alternative



Why the
Change in
Recent
Years?

Open-Source Model=No Licensing Costs

Completely free and open source
(PostgreSQL License=liberal BSD/MIT-style license)

Unlike Oracle, SQL Server and other enterprise DBs:

- No per-core licensing
- No feature gating
- No **audit fears** or surprise costs.

A desire to redistribute budgets to cloud and AI innovation.

The Ecosystem Explosion

PostgreSQL
isn't just
one thing:

Over 25+ PostgreSQL compatible
flavors

Why this
matters:

Horizontal scalability

Multi-model support

Serverless

Time-series

JSON-native capabilities

Massive Feature Growth

It's not the sleepy RDBMS of the 2000's.

150+ major features added over last 5 releases, including:

Regular major releases every year, LTS support for 5+ years

Community-Driven Innovation

Global contributor base, not single-vendor project.

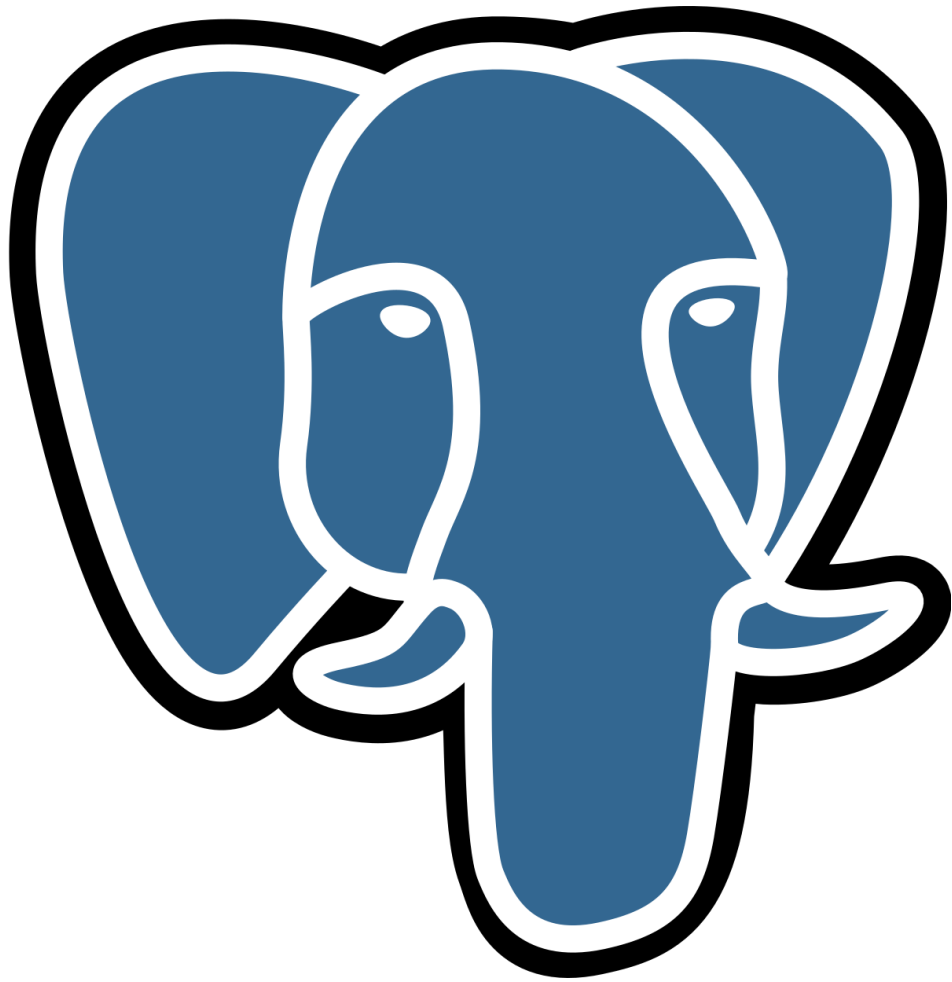
700+ contributors from companies like:

- Microsoft, Redgate, AWS, EDB, Fujitsu, Crunchy Data

Active mailing lists, commit fests and the PostgreSQL Global Development Group as a guiding, but open force.

Cloud and Kubernetes Native

- ▶ PostgreSQL is the default RDBMS for Kubernetes-native apps
 - ▶ Crunchy Postgres Operator
 - ▶ CloudNativePG Operator
- ▶ Integrated with GitOps, Helm charts , cloud backups
- ▶ Serverless PostgreSQL:
 - ▶ Neon, Supabase, Aiven, Neon.tech, Nile



The Technical Details

Let's Start with SQL Compliance

- ▶ Full ACID compliance ensure data reliability
- ▶ Support complex queries, window functions, CTEs, JSON/JSONB, arrays, etc.
- ▶ Conforms closely to SQL standards, which makes migrations from many commercial RDBMS platforms more predictable.

They Who Gets All the Data WINS.

- PostgreSQL can handle relational, JSONB and full-text search in a single query.

```
CREATE TABLE products (  
  id SERIAL PRIMARY KEY,  
  name TEXT,  
  attributes JSONB,  
  description TEXT  
);
```

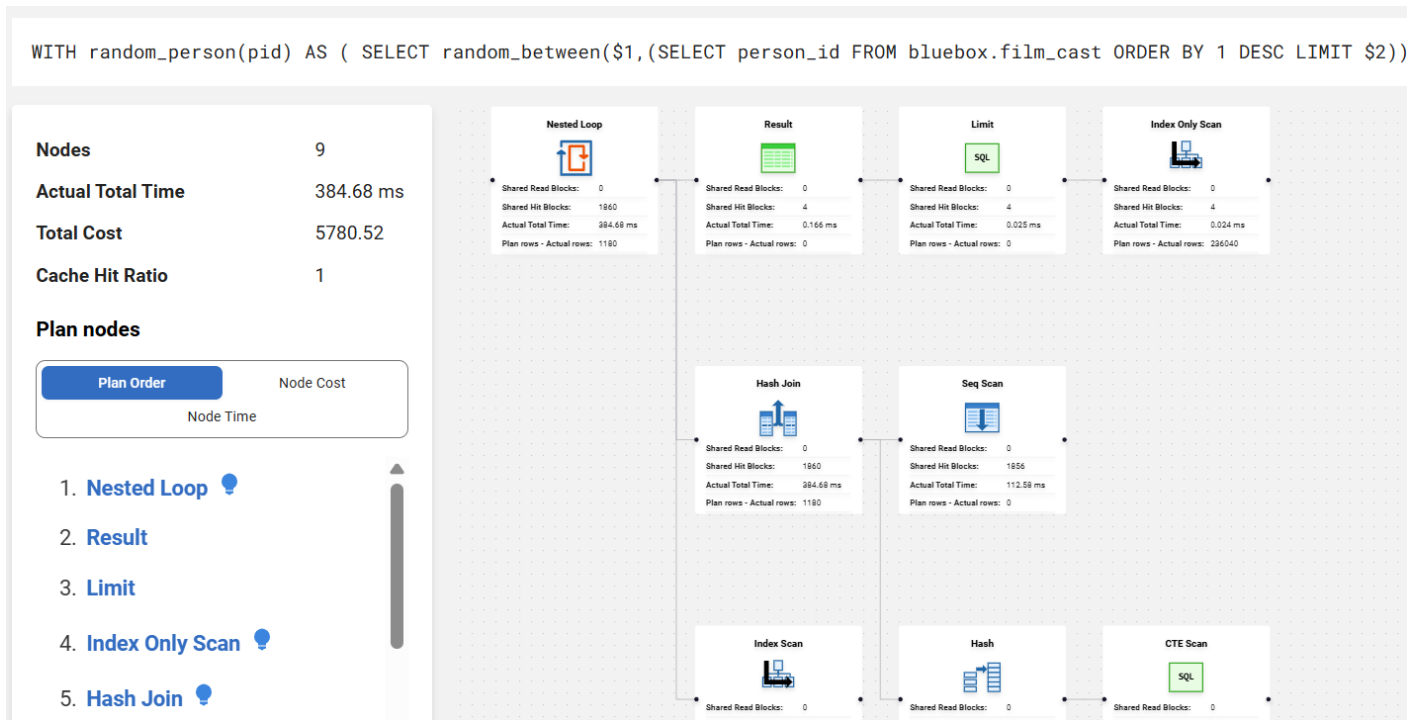
```
INSERT INTO products (name, attributes, description) VALUES  
(  
'Smartphone', '{"brand": "Acme", "specs": {"ram": "8GB", "storage": "128GB"}}',  
'Laptop', '{"brand": "TechCo", "specs": {"ram": "16GB", "storage": "1TB"}}', 'A
```



-- Query JSON, relational, and text search together

```
SELECT name  
FROM products  
WHERE attributes->>'brand' = 'Acme'  
      AND to_tsvector(description) @@ to_tsquery('camera');
```

Performance Data, too



High Availability and Disaster Recovery

- ▶ Streaming Replication is built-in
 - ▶ Asynchronous or synchronous replication from a primary to one or more replicas
 - ▶ A replica can be promoted manually in case of failure.
- ▶ Logical Replication
 - ▶ Available since PostgreSQL 10
 - ▶ Uses a publish/subscribe model
 - ▶ No DDL replication and has some latency.
- ▶ Patroni(with etcd/Consul)
 - ▶ Popular HA framework that wraps streaming replication
 - ▶ Fully automated failover and recovery
- ▶ Repmgr offers semi-automated failover
 - ▶ Requires some scripting
- ▶ pg_auto_failover
 - ▶ Microsoft developed for Citus
 - ▶ Automatic and easier to deploy

HA Continued

- ▶ Kubernetes Operators extend HA
 - ▶ CrunchyData, Zalando's PostgreSQL Operator, StackGres, KubeDB, etc.
 - ▶ These combine streaming replication, automated failover, backup and monitoring in a Kubernetes-native form
 - ▶ Declarative PostgreSQL HA Clusters
 - ▶ They are tied to Kubernetes, though
 - ▶ DevOps expertise is required.
- ▶ Cloud Vendor Managed Service HA
 - ▶ AWS
 - ▶ RDS Multi-AZ, Aurora PostgreSQL(auto failover, replicas)
 - ▶ Azure
 - ▶ Flexible Server with Zone Redundance or Hyperscale Citus
 - ▶ GCP
 - ▶ Cloud SQL HA or AlloyDB with enterprise-grade failover

Backup and Recovery Options

▶ SQL Dump

- ▶ Tool: pg_dump and pg_dumpall
- ▶ Exports schema and data as SQL statements or custom formats
- ▶ Not good for large databases
- ▶ No PIT recovery

```
pg_dump -Fc -U postgres -f mydb.dump mydb
pg_restore -d newdb mydb.dump
```

▶ File System Level Backup

- ▶ Tool: pg_basebackup
- ▶ Creates a consistent snapshot of the data directory.
- ▶ Supports full backup and can include WAL files.
- ▶ Requires full restore of db cluster.
- ▶ Requires downtime if not used with replication or WAL archiving.

```
pg_basebackup -D /pgbackup -Ft -z -P -U replicator
```


Additional Backup and Recovery Options

▶ Continuous Archiving with PITR

- ▶ Tools: WAL Archiving + `restore_command`
- ▶ This is a combination of the base backup and WAL (write-ahead log) archiving.
- ▶ Enables fine-grained recovery, (RPO close to 0)
- ▶ Significant storage required to house WAL files.

▶ pgBackRest

- ▶ Popular 3rd Party Tool for enterprise backups.
- ▶ Supports full, incremental, differential backups, compression, encryption, retention policies, parallel restore and remote storage.
- ▶ Highly performant, even for large databases.
- ▶ Complex to configure.

Extensibility and Customization

- ▶ PostgreSQL allows custom data types, operators, index types (GiST, GIN, SP-GiST, BRIN)
- ▶ Procedural languages and foreign data wrappers (FDWs)
- ▶ Write your own extensions, making it more like a data platform than a static database.
- ▶ Extensions include:
 - ▶ postGIS for geospatial data
 - ▶ pg_partman for time-based partitioning
 - ▶ pg_stat_statements for performance tuning

MySQL Hit to Wordpress Lock-in

- ▶ With installation of PG4WP (PostgreSQL for WordPress, PostgreSQL can be used with a Wordpress site over MySQL or MariaDB.
- ▶ This is now available for many website platforms from the admin console.
- ▶ Removes additional support demands of MySQL, but not the phpadmin management.

Rich Data Type Support

Native support for arrays, hstore, UUIDs, JSON/JSONB, XML, full-text search and geospatial types.



This flexibility enables hybrid use cases, including document store, key-value, relational, etc. all in a single engine!

MVCC (Multi-Version Concurrency Control)

Ensures high-performance, non-blocking reads while maintaining consistency

Excellent for high-concurrency OLTP workloads.

Parallelism and Performance Optimization

Supports parallel query execution, incremental sorting, JIT compilation and table partitioning.

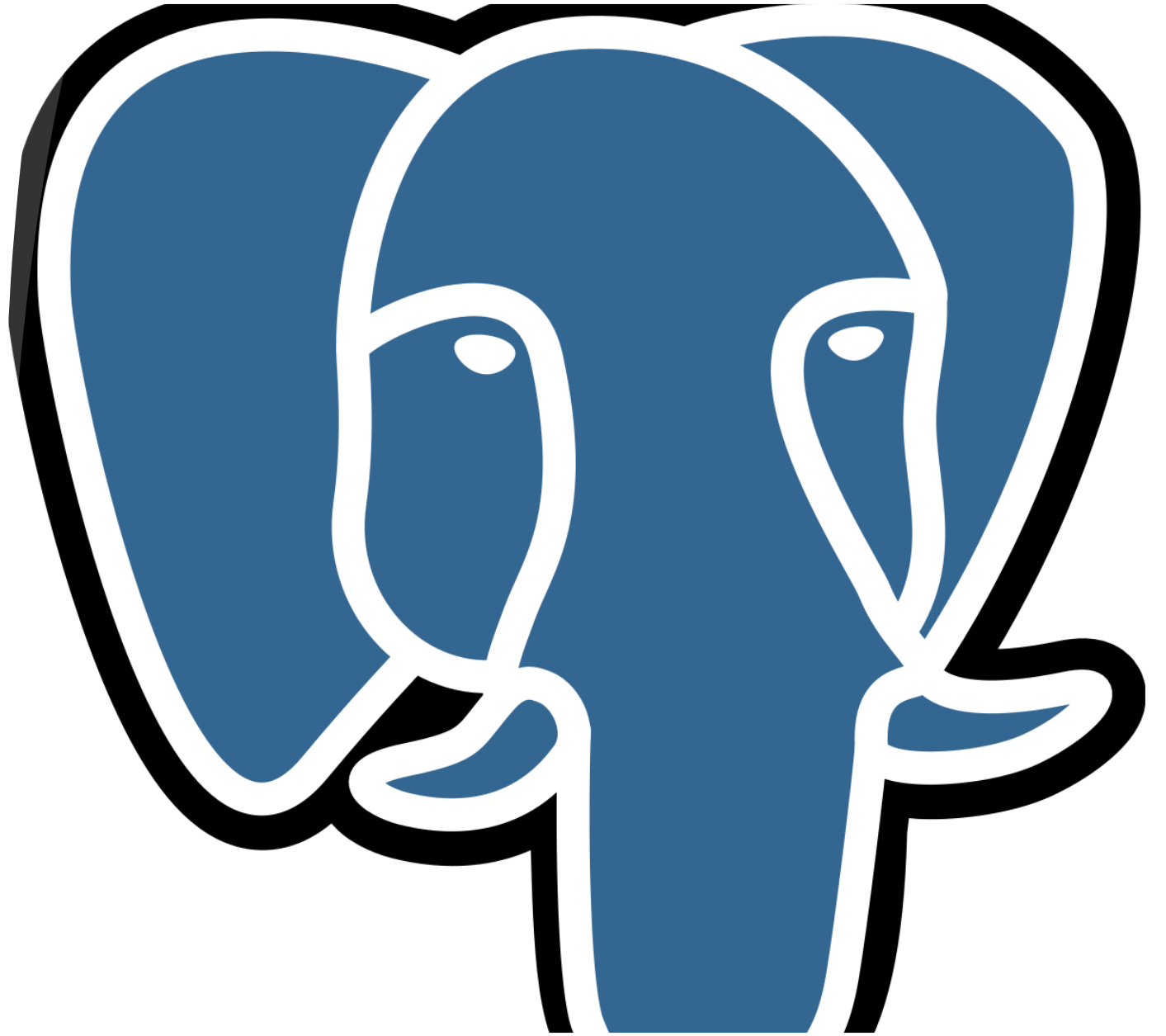
Advanced indexing strategies (i.e. B-tree, GIN, GiST) to improve read performance across various data models.

Security and Role Management

Fine-grained role-based access control (RBAC), role level security, SSL, and encryption.

Supports LDAP, Kerberos, Oauth, and certificate-based authentication.

The More
of
PostgreSQL



Cloud-Native and Ecosystem- Friendly

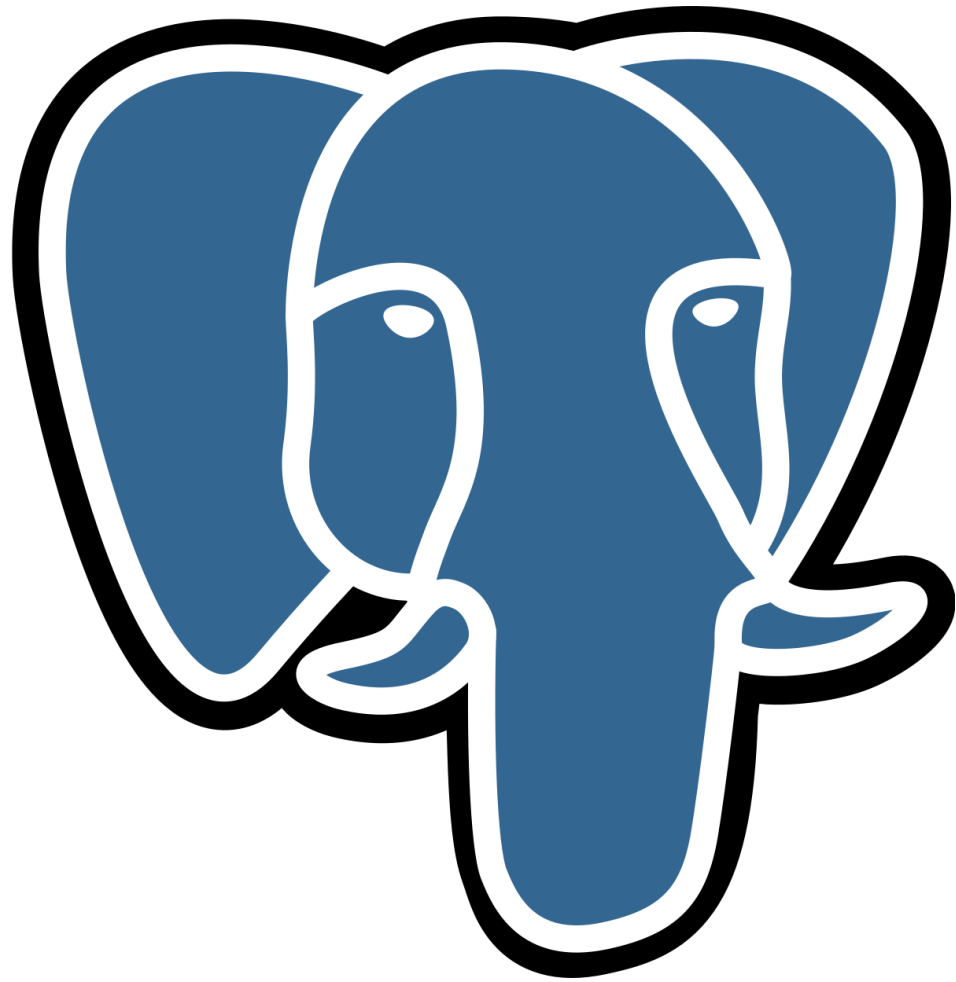
- ▶ Cloud vendors like AWS, Azure and GCP have embraced it because:
 - ▶ Open-source licensing
 - ▶ Multi-tenant readiness
 - ▶ Broad compatibility with modern analytics and AI platforms

Let's Talk Foreign Data Wrappers (FDW)

- ▶ This offers PostgreSQL the ability to query external data sources like Oracle, SQL Server, MongoDB, CSV files, REST APIs and even other PostgreSQL instances, enabling data federation and integration.

Let's Not
Forget...

- ▶ Vibrant Community and Innovation
 - ▶ PostgreSQL Global Development Group and an active open-source community release major updates every year, with long-term support.
 - ▶ Contributions are often driven by real-world use cases.
 - ▶ Ideas stem from many large-scale deployments.



The Wins and the Opportunities

The Win: Instagram/Meta with PostgreSQL

Why This Matters

- Meta (Instagram) chose PostgreSQL for critical, high-performance workloads,
- Demonstrates PostgreSQL's capability to scale to billions of users while maintaining consistency and availability

Key Benefits Realized

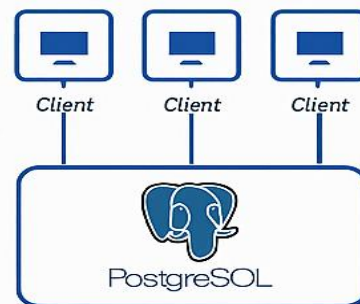
- Performance at Scale: Handles massive read/write traffic with low latency
- Flexibility: Extensible data types and indexing to optimize workloads
- Cost Efficiency: Open-source, eliminating licensing costs of proprietary databases

Takeaway: If PostgreSQL can power Instagram's global workloads, it can handle your mission-critical applications.



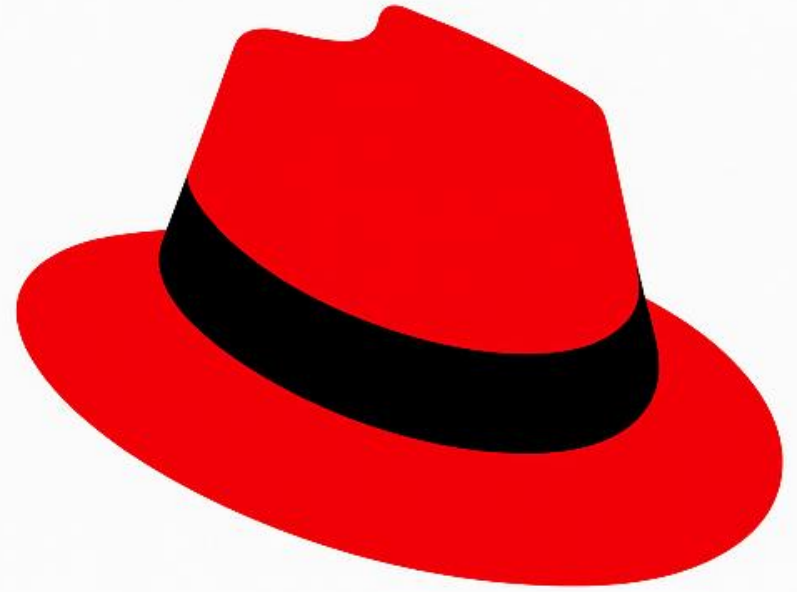
The Impact

- Instagram processes millions of transactions per second on PostgreSQL
- PostgreSQL supports multiple core workloads, including feeds, analytics, and user data



Red Hat:

PostgreSQL powering
internal systems and
analytics



Why PostgreSQL for GitLab?

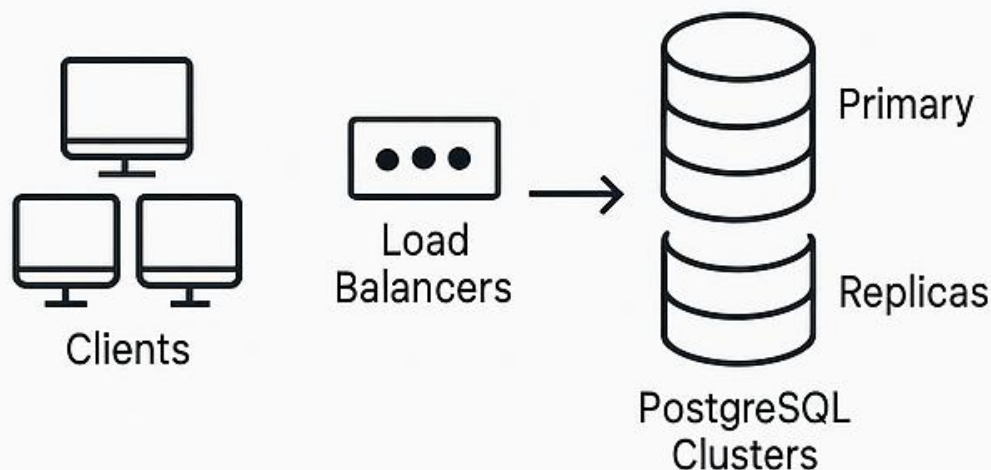
- Open-source, reliable, and feature-rich relational database
- Excellent support for high-concurrency workloads
- Flexibility with extensions for custom needs

How GitLab Scaled It:

- Horizontal scaling with *Patroni* and *PgBouncer* for HA and connection pooling
- Sharding & partitioning strategies to handle millions of repositories and CI/CD pipelines



GitLab



Supports **tens of millions** of projects and pipelines globally



Takeaway: GitLab's success demonstrates how PostgreSQL can scale to power large, complex, cloud-native platforms.

NASA

PostgreSQL behind
telemetry and
research data
systems



Zoological Society of London (ZSL): AI + PostgreSQL + BigQuery on Google Cloud



Why This Matters

- Conservation of 600 wild Asiatic lions in Nepal's habitats
- Combines AI, PostgreSQL, and BigQuery to track populations, health and movement
- Cloud-powered analytics improve response times and conservation decisions.



PostgreSQL



BigQuery



PostgreSQL vs. Main RDBMS

Feature	PostgreSQL	Oracle	SQL Server	MySQL
Licensing	Free	Expensive	Expensive	Free (but limited)
JSON Support	✓ (JSONB, Indexable)	✓	✓	Basic
Extensibility	✓ (FDWs, extensions)	✗	⚠	⚠
Community	Open	Closed	Vendor-led	Dual (Oracle)
Cloud Support	Universal	Limited (OCI)	Azure-only native	AWS/GCP mostly

Demo Time

What this Demo Covers:

- JSONB Storage & GIN Indexing
- pg_trgm Extension for similarity search
- Modern indexing strategies
- Verify in comparison

What You Can Do Today



OLTP



OLAP: Citus, Greenplum



Time-series: TimescaleDB



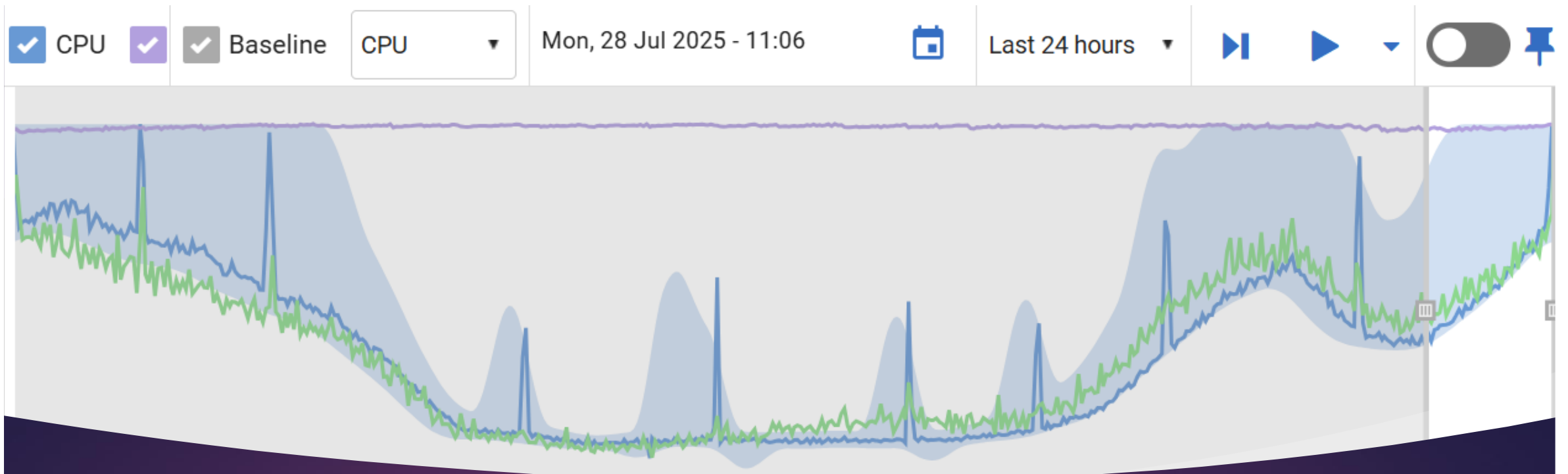
Federated/Polyglot: FDWs



Event-driven: LISTEN/NOTIFY



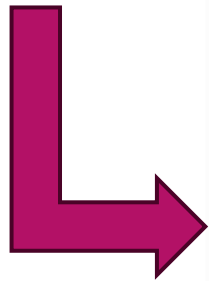
JSON-based document stores



Strategically Monitor and Alert

- ▶ Combination of OS level stats and internal database performance information.
- ▶ It's in its infancy, but shows resource usage, wait events types(categories) explain plans and vital performance information.
- ▶ Additional performance data can be collected from `pg_stat_activity` and various views/extensions.

```
EXPLAIN ANALYZE
SELECT * FROM orders o
JOIN customers c ON o.customer_id = c.id
WHERE o.order_date > now() - interval '30 days';
```



```
Nested Loop (cost=0.84..125.45 rows=10 width=256) (actual time=0.032..1.215 rows=
-> Index Scan using idx_orders_order_date on orders o (cost=0.42..30.12 rows=
    Index Cond: (order_date > (now() - '30 days'::interval))
-> Index Scan using customers_pkey on customers c (cost=0.42..9.54 rows=1 wid
    Index Cond: (id = o.customer_id)
Planning Time: 0.255 ms
Execution Time: 1.235 ms
```

Full Explain Plans

Missing Performance Data

- ▶ No canned reports, (like AWR/ASH/ADDM in Oracle)
- ▶ Significant data still at host level
- ▶ No session level information, like Oracle's ASH data or SQL Server's sys.dm_exec_wait_stats.
- ▶ Requires third party tools to monitor historical data
- ▶ Limited query execution information and requires pg_store_plans for history.
- ▶ Very little to no I/O latency breakdown.

Overall Challenges and Considerations

PostgreSQL can be CPU-heavy on complex queries

Vacuuming and tuning still requires platform expertise

Not all features are plug-and-play, (i.e. partitions)

Scale-out requires third-party extensions: Citus, Yugabyte

Summary – Why PostgreSQL is Winning



No license cost, rich features and community driven



Suitable for modern cloud-native, hybrid and on-prem workloads



Faster innovation cycle than traditional enterprise RDBMS'



Support from hyperscalers, startups and major enterprises alike.

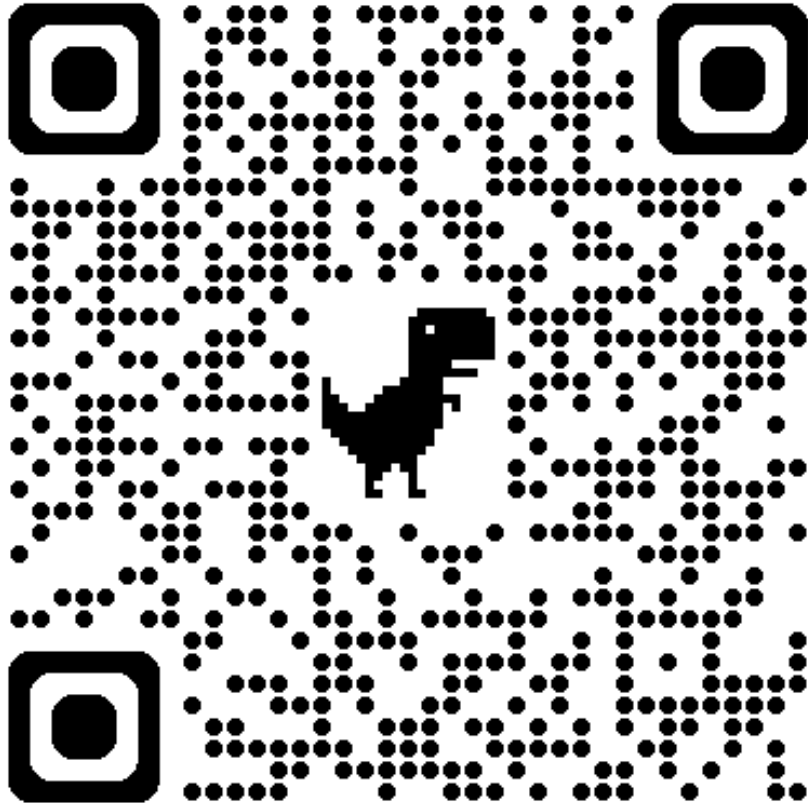
Want to Learn/Do More?

- ▶ Start a test project
- ▶ Compare with your existing workload
- ▶ Explore these examples:
 - ▶ Supabase (development friendly)
 - ▶ Crunchy Data or Aiven (enterprise)
 - ▶ Azure DB for PostgreSQL, RDS for Aurora, GCP Cloud SQL (cloud ready)
 - ▶ Heck, you can run Google's AlloyDB Omni ANYWHERE....

References

- ▶ PostgreSQL Downloads: <https://www.postgresql.org/download/>
- ▶ Supabase Download: [GitHub - supabase/supabase: The Postgres development platform. Supabase gives you a dedicated Postgres database to build your web, mobile, and AI applications.](#)
- ▶ AlloyDB Omni: [AlloyDB Omni | Google Cloud](#)
- ▶ 1000 PostgreSQL Projects: [PostgreSQL Projects - 1000 Projects](#)
- ▶ Postgres Related Projects: [PostgreSQL: Related Projects](#)
- ▶ Github PostgreSQL Projects: [GitHub - Sarohaa199/PostgreSQL-Projects: This is PostgreSQL projects for data analytics](#)
- ▶ Redgate Free tools, pgNow and pgCompare: <https://www.redgate.com/products/redgate-pgnow/installer/> and <https://www.redgate.com/products/redgate-pgcompare/trial/> (choose community)

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Q&A

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